Tarteel-ML Dataset Exploration Report

DSAI 353 Lab 11: Tarteel-ML Dataset Exploration

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Contents

1 Overview of the Tarteel Dataset and Its Purpose

The Tarteel dataset is a comprehensive collection of Quranic audio recordings designed to facilitate the development of machine learning models for Quranic audio analysis. The dataset aims to advance Islamic technological tools and support Quran learning by providing a diverse set of recordings from users with varied backgrounds and devices. Its primary purpose is to enable researchers and developers to create applications such as recitation recognition, tajweed error detection, and voice-based Quranic search, thereby enhancing accessibility and educational resources for Quranic studies.

2 Data Structure: Content and Organization

The Tarteel dataset is meticulously organized to support machine learning tasks. Key aspects of its structure include:

- **Metadata Format**: Stored in JSON or CSV format, containing essential recording information.
- **Audio Format**: Audio files are saved as WAV files with varying sample rates, suitable for audio processing.
- **Key Fields**: Includes surah number, ayah number, recording ID, timestamp, and device information.
- **Organization**: Structured by surah and ayah numbers for easy access and analysis.
- **Storage**: Hosted on an AWS S3 public bucket, ensuring accessibility for researchers.

This organization facilitates efficient data retrieval and preprocessing for machine learning applications.

3 Sample Visualizations

The exploration of the Tarteel dataset includes several visualizations to understand its composition and characteristics. Below are the key visualizations generated from a synthetic sample of 500 recordings.

3.1 Distribution of Surahs in the Dataset

The distribution of surahs in the dataset highlights which surahs are most frequently recorded. Figure 1 shows a bar plot of the top 20 surahs, indicating their prevalence in the synthetic dataset.

3.2 Distribution of Ayahs in the Dataset

Similarly, the distribution of ayahs provides insights into the verses most commonly recited. Figure 2 presents a bar plot of the top 20 ayahs, showcasing their frequency.

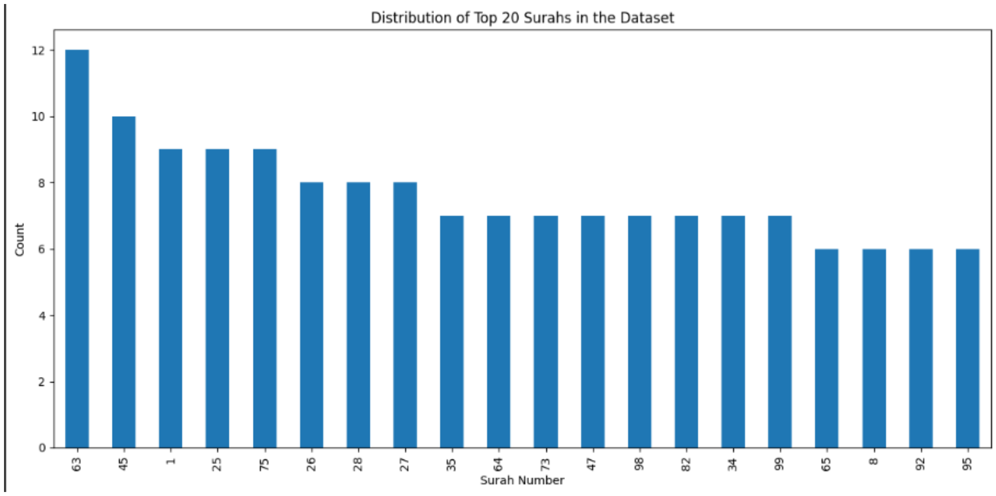

Figure 1: Distribution of the top 20 surahs in the Tarteel dataset.

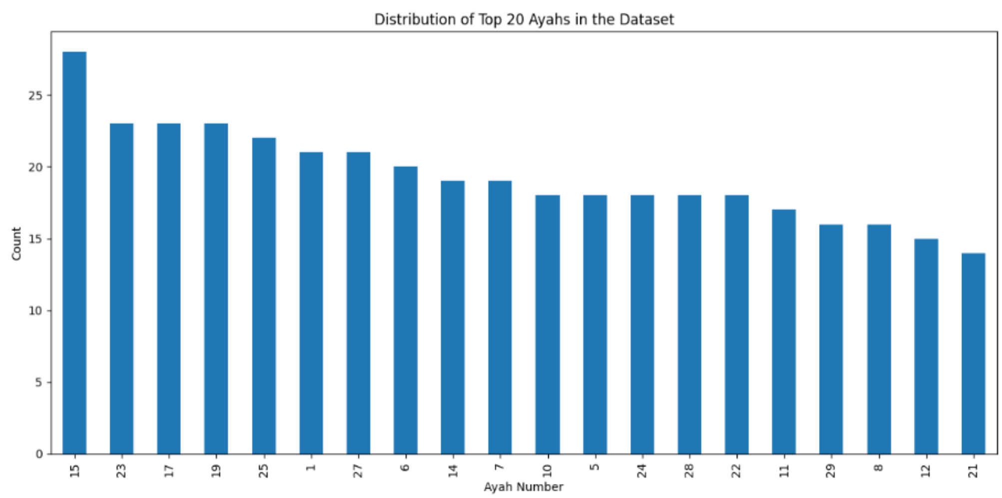

Figure 2: Distribution of the top 20 ayahs in the Tarteel dataset.

3.3 Recording Device Distribution

The dataset includes information on the devices used for recordings, which is critical for understanding recording quality and variability. Figure 3 displays a pie chart of the device distribution, highlighting the prevalence of devices such as iPhones and Android phones.

3.4 Audio Waveforms, Spectrograms, and MFCCs

Three sample recordings were analyzed to visualize their audio characteristics, including waveforms, spectrograms, and Mel-frequency cepstral coefficients (MFCCs). Figures 4, 5, and 6 present these visualizations for three synthetic recordings.

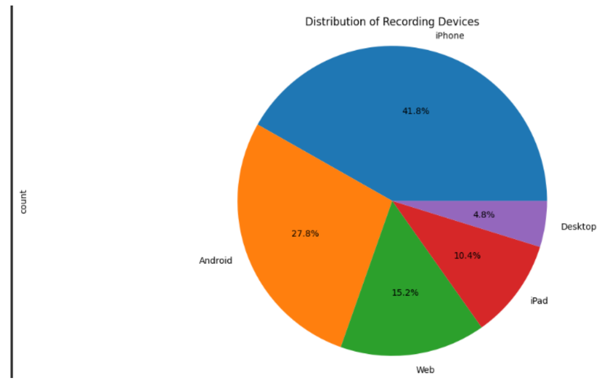

Figure 3: Distribution of recording devices in the Tarteel dataset.

4 Preprocessing Steps Needed Before Model Training

To prepare the Tarteel dataset for machine learning model training, several preprocessing steps are essential:

- **Resampling**: Standardize sample rates to 16 kHz to ensure consistency across recordings.
- **Silence Trimming**: Remove silent segments to focus on meaningful audio content.
- **Normalization**: Adjust volume differences to normalize audio amplitudes.
- **Feature Extraction**: Extract features such as MFCCs and Mel spectrograms for model input.
- **Data Augmentation**: Apply techniques like time stretching and pitch shifting to enhance dataset robustness.
- **Train/Validation/Test Splitting**: Divide the dataset into training, validation, and test sets for model evaluation.

Figure 7 illustrates these preprocessing steps applied to a sample recording, showing the original waveform, resampled waveform, trimmed waveform, normalized waveform, and MFCC features.

5 Use Cases: Potential ML/NLP/Audio Projects

The Tarteel dataset supports a variety of machine learning, natural language processing, and audio processing projects. The following use cases were identified:

1. **Quranic Recitation Recognition**: Build a model to identify the surah and ayah being recited, using techniques such as CNNs, RNNs, LSTMs, and Transformer models. Applications include educational tools and automated recitation assessment.
2. **Tajweed Error Detection**: Develop a system to detect tajweed errors in recitations, employing audio classification and sequence-to-sequence models. This can assist learners and provide automated feedback.

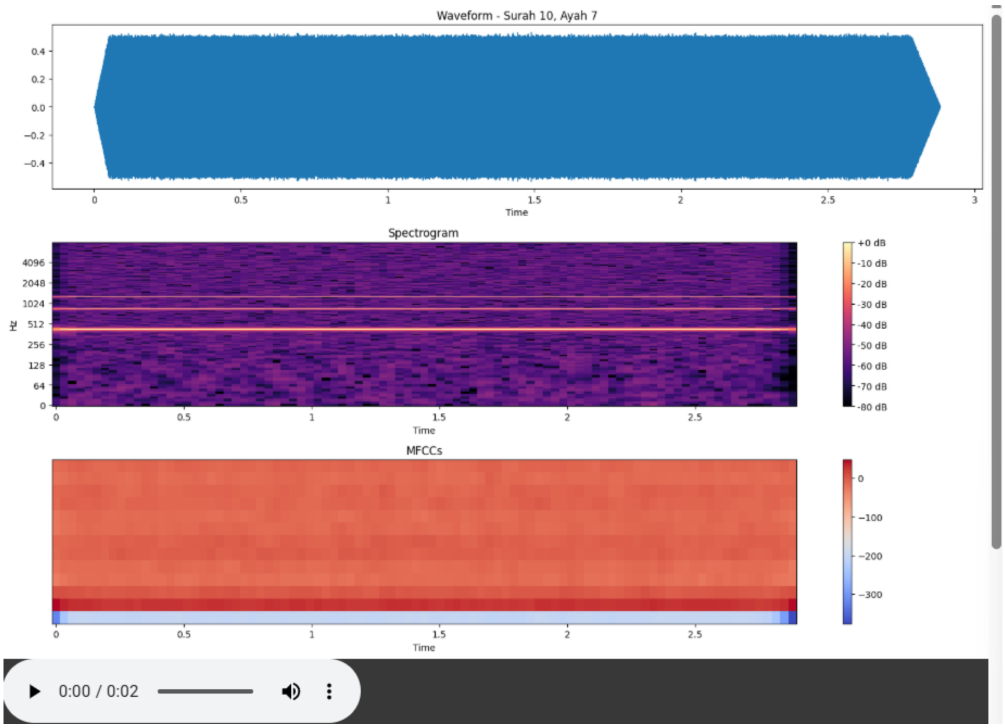

Figure 4: Audio analysis for synthetic recording 1: waveform, spectrogram, and MFCCs.

3. **Voice-based Quranic Search**: Create a system where users recite part of an ayah to find the complete verse, using audio fingerprinting and embedding similarity. This enhances Quran study tools and accessibility.

4. **Recitation Style Classification**: Classify different recitation styles and maqamat using transfer learning and feature extraction, supporting cultural preservation and educational resources.

5. **Recitation Quality Assessment**: Automatically assess recitation quality using regression models and quality metrics, aiding self-improvement and teaching.

6 Proposed Project: Quranic Recitation Recognition

6.1 Problem Statement

The goal is to develop a machine learning model that accurately identifies the surah and ayah being recited from audio recordings, enabling applications such as automated recitation verification and educational tools for Quran learning.

6.2 Methodology

The project will involve the following steps:

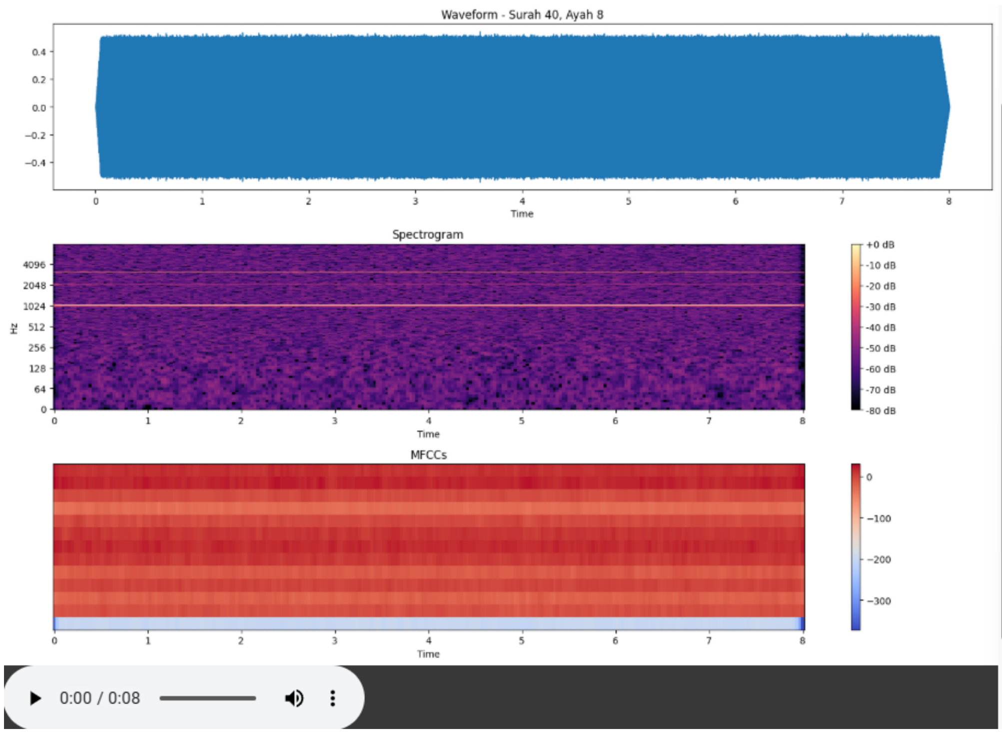

Figure 5: Audio analysis for synthetic recording 2: waveform, spectrogram, and MFCCs.

- **Data Preparation**: Preprocess the Tarteel dataset by resampling audio to 16 kHz, trimming silence, normalizing, and extracting MFCC features.
- **Model Architecture**: Design a hybrid CNN-LSTM model to capture both spectral and temporal features of the audio.
- **Training**: Train the model on the preprocessed dataset, using a train/validation/test split of 70%/15%/15%.
- **Evaluation**: Assess model performance using accuracy and F1-score metrics.

6.3 Expected Outcomes

The model is expected to achieve high accuracy in identifying surahs and ayahs, supporting real-time applications in educational settings and mobile apps.

6.4 Technical Approach

- Use Librosa for audio preprocessing and feature extraction.
- Implement the model using TensorFlow or PyTorch.
- Optimize hyperparameters using grid search and cross-validation.

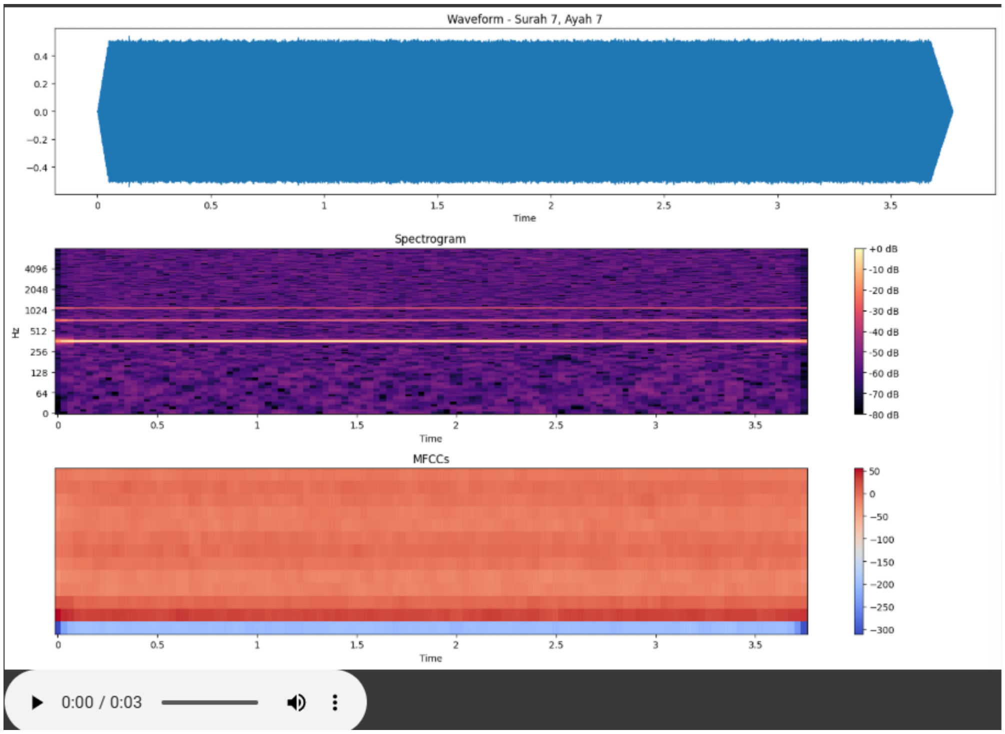

Figure 6: Audio analysis for synthetic recording 3: waveform, spectrogram, and MFCCs.

6.5 Evaluation Metrics

- **Accuracy**: Percentage of correctly identified surahs and ayahs.
- **F1-Score**: Balances precision and recall for robust evaluation.
- **Confusion Matrix**: Analyzes misclassifications across surahs and ayahs.

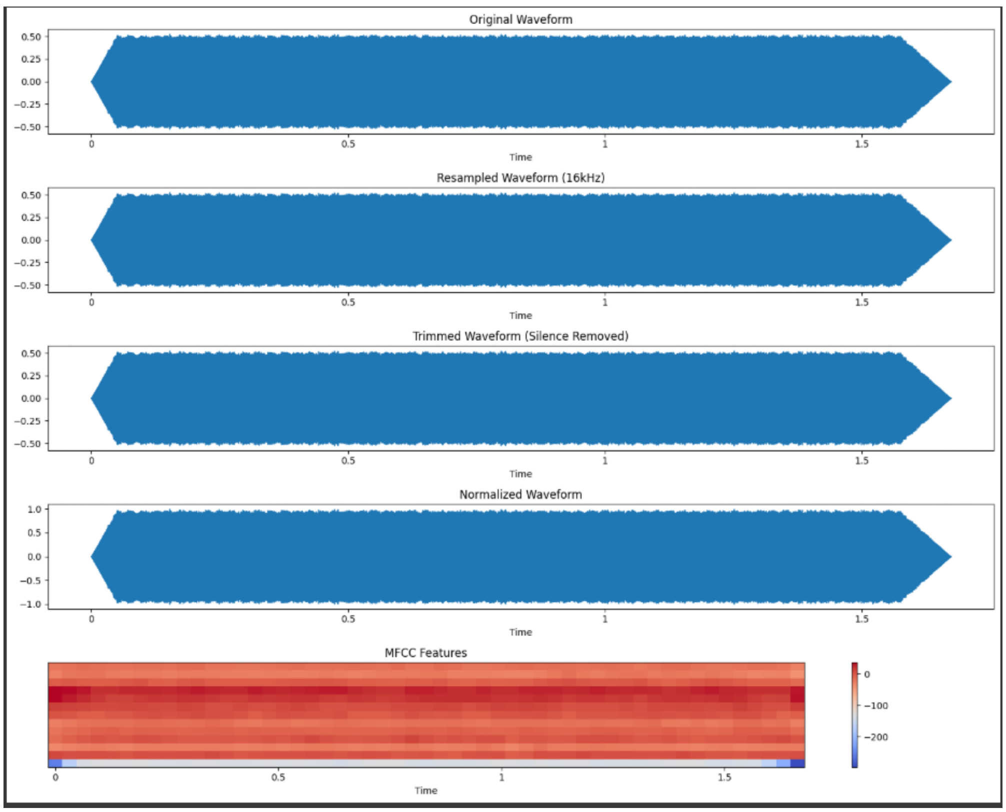

Figure 7: Preprocessing steps for a sample audio recording: original, resampled, trimmed, normalized, and MFCC features.